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Automated Retrosynthesis Planning of Macromolecules Using Large Language Models and Knowledge Graphs

*Qinyu Ma, Yuhao Zhou, and Jianfeng Li**

Identifying reliable synthesis pathways in materials chemistry is a complex task, particularly in polymer science, due to the intricate and often nonunique nomenclature of macromolecules. To address this challenge, an agent system that integrates large language models (LLMs) and knowledge graphs is proposed. By leveraging LLMs' powerful capabilities for extracting and recognizing chemical substance names, and storing the extracted data in a structured knowledge graph, the system fully automates the retrieval of relevant literature, extraction of reaction data, database querying, construction of retrosynthetic pathway trees, further expansion through the retrieval of additional literature and recommendation of optimal reaction pathways. By considering the complex interdependencies among chemical reactants, a novel Multi-branched Reaction Pathway Search Algorithm (MBRPS) is proposed to help identify all valid multi-branched reaction pathways, which arise when a single product decomposes into multiple reaction intermediates. In contrast, previous studies are limited to cases where a product decomposes into at most one reaction intermediate. This work represents the first attempt to develop a fully automated retrosynthesis planning agent tailored specially for macromolecules powered by LLMs. Applied to polyimide synthesis, the new approach constructs a retrosynthetic pathway tree with hundreds of pathways and recommends optimized routes, including both known and novel pathways.

1. Introduction

Retrosynthesis planning[1,2] plays a pivotal role in chemistry research by systematically designing synthetic pathways for target compounds. This approach involves breaking down complex molecules into simpler precursors through a stepwise reverse analysis, ultimately leading to a synthesis plan using commercially available starting compounds. By systematically evaluating potential chemical transformations, retrosynthesis enables researchers to explore a wide range of synthetic possibilities and select the most effective routes. Beyond its direct applications in organic synthesis, retrosynthesis is instrumental in accelerating material discovery, optimizing chemical production processes, and driving innovative research across multiple scientific disciplines.

Conventional rule-based or expert-based computer-aided synthesis[3,4] has obvious limitations, such as high labor costs and limited search space. In recent years, dramatic breakthroughs driven by deep learning have revolutionized retrosynthesis. Single-step retrosynthesis using deep learning, which involves determining reactants based on a given product, is mainly categorized into template-based methods,[5,6] which rely on predefined reaction templates for high precision but have limited applicability, and template-free methods,[7,8] which offer greater flexibility but often sacrifice precision. These retrosynthetic analyses have predominantly focused on small molecules, with limited attention to macromolecules such as polymers and proteins. Multi-step retrosynthesis, which involves deriving a complete reaction pathway from the target product to commercially available starting compounds, can be achieved through search algorithms like proof-number search,[9,10] Monte Carlo Tree Search (MCTS)[2,7] and A* Search.[5,6] However, these techniques primarily focus on decomposing target compounds into one intermediate and multiple starting compounds, leaving more complex multi-intermediate pathways largely unexplored.

The challenges in applying retrosynthesis planning to macromolecules are particularly noteworthy. Unlike small molecules, macromolecules often lack extensive, well-documented reaction databases, making the use of deep learning models for prediction tricky. Moreover, the large number of atoms in macromolecular systems, as well as the fact that chemical reactions are often influenced by complex interactions, make accurate calculations challenging. Therefore, researchers are often required to browse a large amount of academic papers for retrosynthesis planning of macromolecules. Unfortunately, the extraction of reaction information from the literature and the construction of retrosynthetic pathways for macromolecules is further complicated by their complex and variable nomenclature, which makes traditional rule-based methods insufficient for accurately identifying relevant reactions. For instance, the polymer widely known as "polystyrene" may also appear as "poly(1-phenylethylene)"

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based on structure-based naming or as "poly(vinylbenzene)" and "poly(ethenylbenzene)" under source-based conventions.[11,12] To address these issues, more intelligent approaches are necessary. A promising solution lies in leveraging Large Languge Models (LLMs) to ensure the consistency of polymer material names, thereby enabling the construction of an entity-aligned knowledge graph[13] to facilitate the automated construction of retrosynthetic pathways. Despite the potential of this approach, no prior studies have investigated the integration of LLMs and knowledge graphs specifically for retrosynthesis planning. While Bran et al.[14] previously utilized LLMs to automate aspects of chemistry research, their work treated retrosynthesis planning as a supporting tool, relying on underlying deep-learning methods for its implementation.

On the other hand, LLMs[15,16] have reshaped natural language processing with their human-like text generation, complex pattern recognition, and adaptability to tasks including translation, summarization, and question answering. Leveraging deep learning, they process vast amounts of text data, proving invaluable for text mining[17–19] research planning [14,20,21] and chemical applications.[22] Recently, several studies have explored the integration of LLMs with knowledge graphs, notably in materials design,[23] automated scientific discovery,[24,25] and advanced reasoning techniques,[26] further expanding their potential in scientific research. Despite these advantages, LLMs face critical limitations.[27,28] Their probabilistic nature[29] and reliance on unverified data can lead to hallucinations,[30,31] while static datasets delay knowledge updates.[32] They also struggle with precise math,[33] logic,[34] and interpreting non-textual data like molecular structures or reaction schemes. In retrosynthesis planning, these limitations are particularly problematic, as the process requires accurate multi-step retrosynthesis, real-time scientific knowledge, and the ability to assess pathway feasibility. To address these challenges, methods like grammar and reinforcement learning, such as RLHF,[35] DPO,[36] GRPO,[37] and PRefLexOR[38] have been employed to help LLMs generate semi-structured outputs, such as JSON and tables. However, these techniques still face challenges when it comes to producing more complex, highly structured outputs, such as graphs or tree-like structures. Recent innovations, such as Tree-of-Thought[39] and Graph-of-Thought[40] have further enhanced LLMs' reasoning abilities. Yet, these methods remain insufficient for handling complex chemical reaction pathways, especially when these pathways involve large numbers of nodes and intricate relationships. Consequently, LLMs still struggle to reliably chart complex chemical pathways, particularly in the case of macromolecules.

To address these challenges, we propose a retrosynthesis planning agent based on LLMs and knowledge graphs for materials chemistry. This agent is capable of automatically querying, downloading, and extracting chemical reaction information based on a given target product (see the demo video in SI). It then constructs a structured knowledge graph, facilitating efficient and accurate information retrieval and expansion. The agent utilizes a Memorized Depth-first Search (MDFS) algorithm,[41–43] along with database queries, to construct a retrosynthetic pathway tree that synthesizes the target product using commercially available compounds as starting compounds. When a reaction pathway cannot be further expanded, the agent automatically retrieves and incorporates additional synthesis data from relevant literature, continuously enriching the knowledge graph and further broadening and extending the chemical reaction pathways. Ultimately, with the help of the Multi-branched Reaction Pathway Search (MBRPS) algorithm, the agent identifies all authoritative and feasible synthesis pathways, including multi-branched ones, by considering the complex relationships among chemical reactants. It then recommends the optimal reaction pathway based on factors such as reaction conditions, yields, and so on. The proposed approach is applied to polyimide synthesis, showcasing its ability to construct complex retrosynthetic pathway trees and recommend optimized routes, encompassing both established and novel pathways.

2. Experimental Section

2.1. Automated Literature Retrieval

The workflow of the automated retrosynthesis planning agent is illustrated in **Figure 1**. A demo video in Supporting Information is also provided, showcasing the agent's ability to execute the entire workflow autonomously without any human intervention. The Agent first utilizes the Google Scholar API[44] to retrieve relevant paper titles based on predefined keywords. These titles are then used to download literature PDFs via web scraping. Text is extracted from the PDFs using PyMuPDF.[45] Following extraction, the data is cleaned by removing special characters and symbols to enhance readability and ensure better comprehension by LLMs.

2.2. Knowledge Graph Construction From Extracted Information

Our agent is developed based on the ChatGPT-4o API.[46] Leveraging this model, the agent integrates techniques such as prompt engineering,[47] in-context learning,[47] and Chain-of-Thought (CoT)[48] to accomplish tasks including entity and relation extraction, knowledge graph construction, and entity alignment as illustrated in **Figure 2**.

It processes the cleaned text and images to extract chemical reactions, which are then output in a standardized format. The extracted chemical reaction information includes the names of reactants and products, reaction temperature, pressure, catalysts, solvents, atmosphere, reaction duration, and yield. Leveraging ChatGPT-4o's inherent proficiency in text comprehension and standardized output, these tasks are effectively accomplished without the need for fine-tuning in most cases.[18,19] However, there are occasional instances where nonreactive reagents are incorrectly listed as reactants, even with the use of prompting to constrain the model. We use the CoT technique to conduct a secondary verification to avoid this issue (See Section S1, Supporting Information).

Based on the model's structured outputs, the agent uses regular expressions to extract entities and relationships. Specifically, each reactant and product is treated as an entity, while reaction conditions, numerical identifiers, and yields are considered as relations, forming unidirectional edges from reactants to products. Ultimately, the agent converts unstructured chemical reaction information from literature into a structured knowledge graph for efficient information retrieval and expansion.

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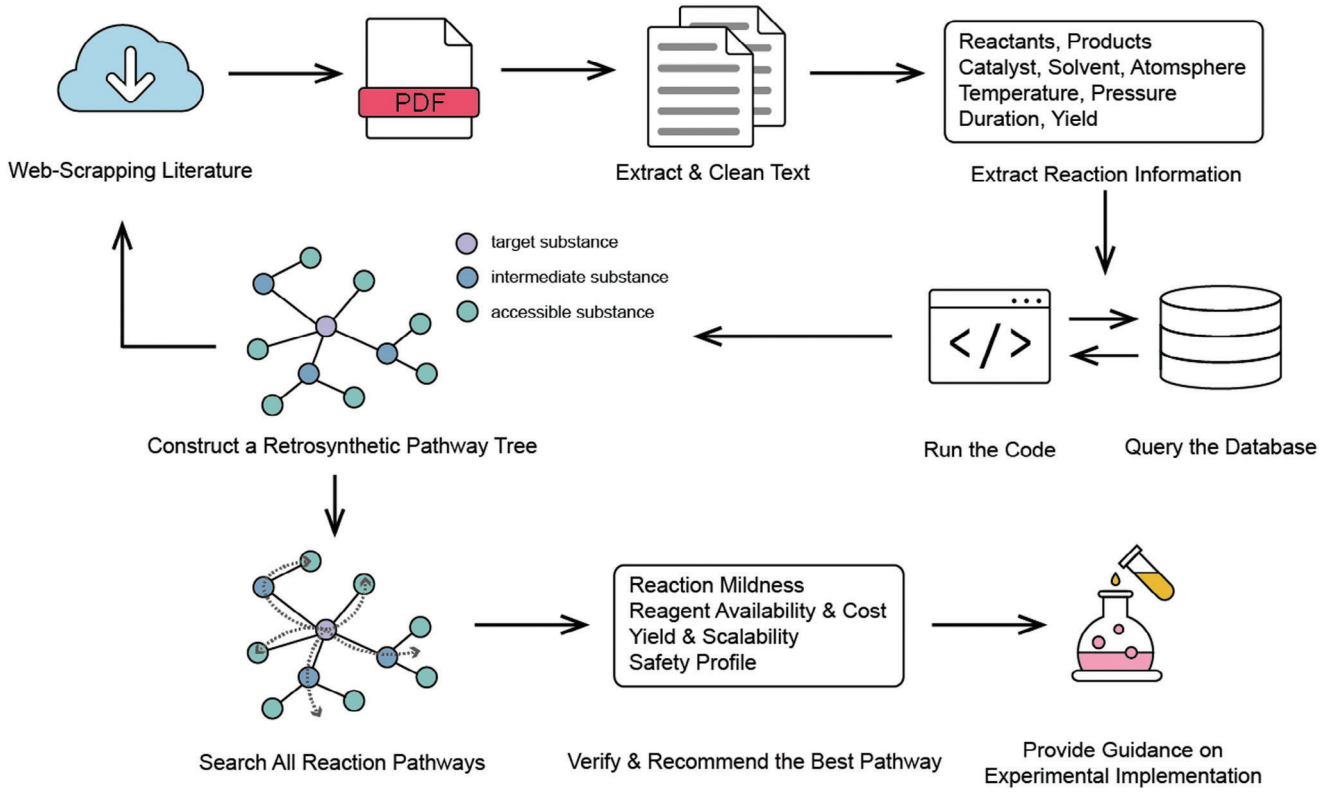

Figure 1. Schematic workflow for automated retrosynthesis planning using the LLM agent, covering literature retrieval, reaction data extraction, database querying, expansion and construction of retrosynthetic tree, and optimal pathway recommendation.

Although prompt engineering and in-context learning can ensure consistency in chemical substance names within a single paper, it is challenging to maintain consistency across multiple papers due to the input length limitations of LLMs. Therefore, the agent rechecks the knowledge graph to identify cases where different nodes represent the same substance. If such cases exist, the agent unifies them and updates the knowledge graph accordingly (See Section S2 Supporting Information).

2.3. Retrosynthetic Pathway Tree Construction and Expansion

Utilizing the constructed knowledge graph, the agent employs a Memorized Depth-first Search (MDFS) algorithm[41–43] to build the retrosynthetic pathway tree, with the target product as the root and leaf nodes representing commercially available compounds.

The goal of constructing a retrosynthetic pathway tree is to trace the reaction pathway step by step from the target substance back to the initial reactants. Specifically, each node in the retrosynthetic pathway tree represents a chemical substance, generated through a specific reaction. The construction of a retrosynthetic tree follows a set of rules:

1) If the target substance is already present in the accessible set of initial reactants, it is marked as a leaf node, requiring no further expansion.
2) If the substance can be synthesized through any known reactions, it is considered expendable; otherwise, if it cannot be synthesized, it is considered non-expandable.

For expandable nodes, the MDFS algorithm traverses all reactions producing the substance, retrieves reactants one by one, and adds them as child nodes to the current node. To prevent cycles, the algorithm discards a path if the new node already exists in the set of parent nodes. This process is carried out recursively, ensuring the validity of each route. Ultimately, only valid reaction pathways are retained, while for nodes that cannot be further expanded or form a cycle, the corresponding reaction pathways will be completely removed, ensuring that the final tree structure accurately reflects the synthesis route from the target substance to the initial reactants.

During the recursive tree construction, the agent queries databases such as eMolecules[49] and PubChem,[50] along with additional commonly used polymers, to verify whether the current node represents a commonly used substance. Additionally, the agent uses RDKit[51] to convert the names of small molecules into standardized SMILES strings for database matching. If a node corresponds to a commonly used substance, it is designated as a leaf node, halting further expansion. To further enhance the efficiency of tree construction, a memory-augmented approach is employed, where the results of database queries, regarding whether a node substance corresponds to commonly used materials, are stored in a cache. This strategy eliminates the need for repeated database lookups of the same substance, significantly reducing computational overhead.

It is worth noting that not all reactants in a single paper are typically commercially available. Therefore, it is necessary to further investigate the literature on the synthesis of intermediate

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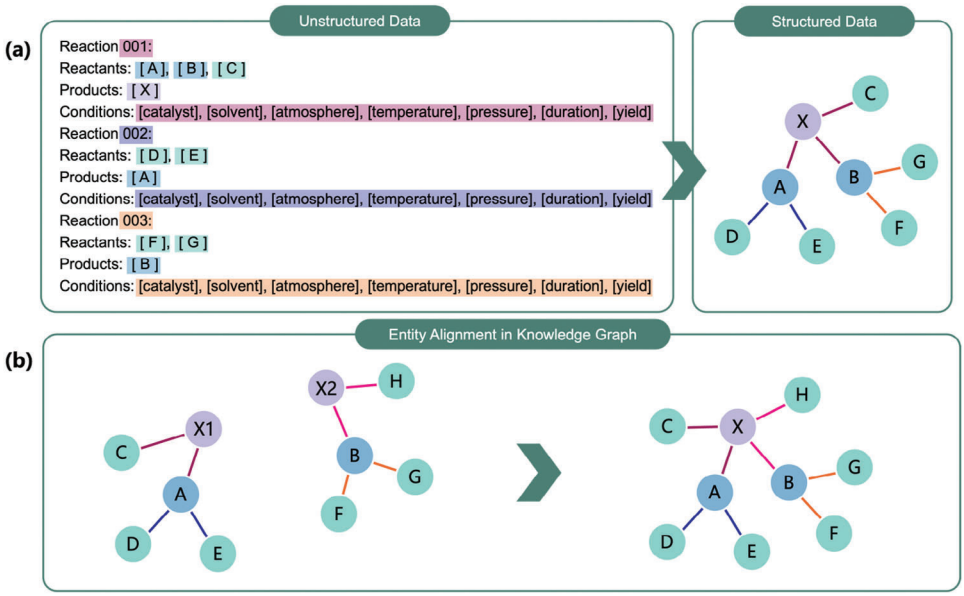

Figure 2. Schematic Diagram of Knowledge Graph Construction. a) Schematic diagram for extracting chemical reactions and converting unstructured data into structured formats. b) Entity alignment in knowledge graphs to ensure consistent naming across articles.

reactants, until commercially available compounds can be used to synthesize the intermediate. Similarly, during retrosynthesis tree construction, if a node cannot be further expanded to a leaf node, the agent will query the relevant literature on the synthesis of the intermediate corresponding to that node, extract relevant chemical reactions from it, and add them to the knowledge graph, thus helping the node successfully expand to a leaf node, enabling the construction of a complete reaction pathway. Ultimately, an expanded retrosynthetic pathway tree with the target substance as the root node is constructed, which includes multiple chemical reaction pathways that can synthesize the target substance from commercially available compounds.

2.4. Chemical Reaction Pathways Search and Recommendation

Upon constructing the retrosynthetic pathway tree for the target product, the agent employs the MBRPS Algorithm, to identify all valid chemical reaction pathways, as illustrated in **Figure 3**. This algorithm is specially designed for multi-branched reaction pathways, which are common in practical retrosynthesis planning (see Discussion section for more details). Each reaction pathway identified provides all the reactions required to synthesize the target product from commercially available compounds, with validation from a database. Specifically, a product can be synthesized through various reactions with corresponding reactants, each representing a node in the retrosynthetic pathway tree. For nodes associated with the same reaction index, they form an "AND" relationship, meaning that all child nodes with the same index must be included to synthesize the target compound. On the other hand, nodes with different reaction indices represent an "OR" relationship, indicating that the target compound can be synthesized by selecting one of several possible reactions. Based on this relationship, we use a recursive method to obtain all reaction paths for each node. This method identifies all valid synthesis routes, including multi-branched ones, helping LLMs overcome weak reasoning in multi-branched paths and enabling comprehensive exploration of reaction pathways.

Finally, the agent evaluates all identified pathways, considering various factors such as the availability and cost of reactants, catalysts, and solvents, the mildness of reaction conditions (e.g., low temperature, pressure, short duration), reaction yield and scalability, and the safety profile of reagents and conditions (e.g., toxicity, hazards), by leveraging CoT. Based on these criteria, the agent recommends the optimal synthetic route for the target product, offering a more efficient and reliable solution for retrosynthesis planning.

3. Result

3.1. Retrosynthetic Pathway Tree of Polyimide

The aforementioned method is applied to polyimide, a high-performance polymer renowned for its exceptional thermal stability, chemical resistance, and mechanical strength. These properties make polyimide indispensable in industries such as aerospace, electronics, and high-temperature applications.[52] However, its complex synthesis and high production costs have driven research into optimizing its synthetic routes.[53] Therefore, we have chosen polyimide for retrosynthesis pathway analysis to explore more efficient and cost-effective approaches. By designating "polyimide" as the target substance for retrosynthetic analysis, the agent retrieved 39 research papers on polyimide synthesis methods, extracted chemical reactions from these sources, and converted them into a structured knowledge graph format, in the first round of the searching process. By integrating database searches, a chemical retrosynthetic pathway tree was recursively constructed.

When the agent encounters an intermediate node that cannot be expanded, it queries about five additional articles on its

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(a)

Algorithm Multi-branched Reaction Pathway Search Algorithm (MBRPS)

```
function SearchReactionPathways(current node)
Require: Current node
Ensure: Reaction pathways as sequences of reaction indices
    if current node is a leaf node then
        return an empty array
    end if
    Initialize PathwaysDict to store reaction pathways for child nodes
    for each child node of the current node do
        Obtain pathways by SearchReactionPathways(child node)
        Get ReactionIdx for the child node
        if ReactionIdx is not in PathwaysDict then
            Add ChildPaths to PathwaysDict under ReactionIdx
        else
            Merge ChildPaths with existing pathways
        end if
    end for
    return pathways in PathwaysDict as an array
end function
```

(b)

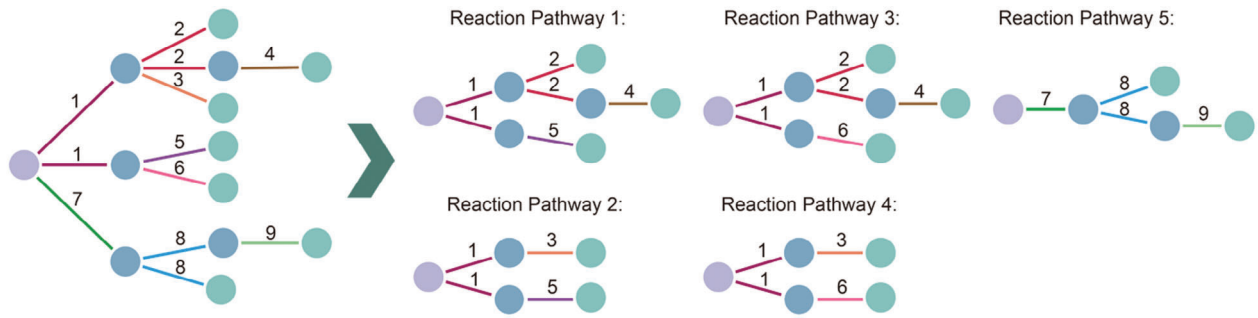

Figure 3. Illustration of the MBRPS algorithm. a) Algorithm pseudocode. b) Example demonstrating the identification of multiple reaction pathways in a retrosynthetic pathway tree.

synthesis methods to extract supplementary chemical reactions, thereby helping to extend the reaction pathway to available compounds as initial reactants. In the end, the agent was supplemented with 158 additional papers on intermediate synthesis reactions, processed a total of 197 papers, and obtained an expanded chemical retrosynthetic pathway tree for polyimide (**Figure 4**). Ultimately, the number of nodes in the Rretrosynthetic Pathway Tree increased from the original 322 to 3099, and the number of synthesis pathways identified through the MBRPS algorithm increased from 55 to 292.

3.2. Evaluation and Recommendation for Chemical Synthesis Pathways

Most studies for retrosynthesis planning focus solely on reactants and products, neglecting reaction conditions.[2,5–8] However, factors such as reaction mildness, reactant availability and cost, yield and scalability, and safety profile are crucial considerations in retrosynthesis planning. Due to a large number of obtained reaction pathways, the agent initially screens reactions within the retrosynthetic pathway tree based on these conditions (see Section S3, Supporting Information for details).

Finally, the agent employs CoT reasoning to conduct a comprehensive evaluation of each reaction pathway that has passed the initial screening and been validated. This evaluation considers each pathway's advantages and disadvantages based on the specific criterion designated by humans. In practical applications, the recommendation criteria can be adjusted based on specific needs. We provide the following two criteria for demonstration purposes:

1) Method for producing commercially available Katpon polyimide.
2) Presence of phenol in the initial reactants.

Based on this detailed evaluation, the agent then recommends the optimal reaction pathway, along with a rationale explaining how it best the outlined criteria. The final recommended reaction pathways are presented in Figure 5 (based on Criterion 1) and Figure 6 (based on Criterion 2) (see Section S4, Supporting Information for details). Notably, the reaction pathway obtained based on Criterion 2 is one of the newly proposed pathways. It was identified by the agent through an extended search of the literature related to intermediate synthesis. This approach enables

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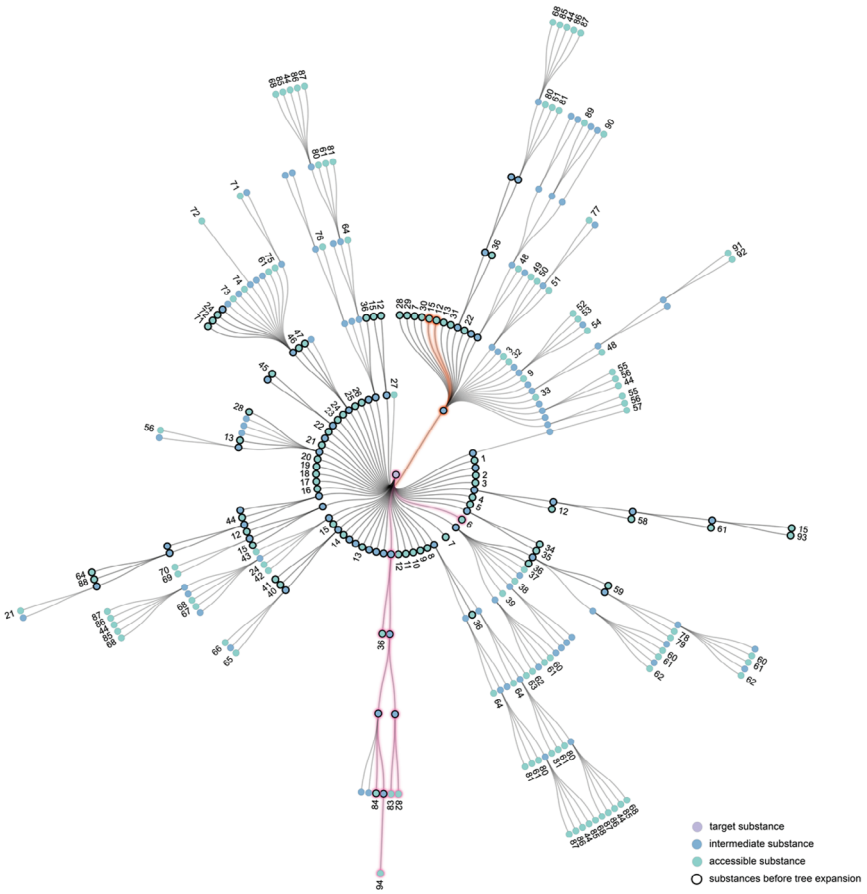

1: 5-amino-1,3,3-trimethylcyclohexanemethylamine	2: 4-chloro-2,5,6-trifluoro-1,3-phenylenediamine	3: 2,2-bis(3,4-dicarboxyphenyl) hexafluoropropane dianhydride
4: 3,3'-diamino-4,4'-dihydroxybiphenyl	5: bis(2,4-pentanedionato)palladium(ii)	6: 4,4'-oxydiphthalic anhydride
7: 3,3',4,4'-biphenyltetracarboxylic dianhydride	8: 3,3',4,4'-biphenyltetracarboxylic acid	9: 4,4'-hexafluoroisopropylidene diphthalic anhydride
10: trans-1,4-diaminocyclohexane	11: 5-(2,5-dioxotetrahydrofuryl)-3-methyl-3-cyclohexene-1,2-dicarboxylic anhydride	12: 4,4'-oxydianiline
13: 3,3',4,4'-benzophenone tetracarboxylic dianhydride	14: 4,4'-diaminodiphenyl methane	15: pyromellitic dianhydride
16: triethylamine	17: 1,4,5,8-naphthalenetetracarboxylic dianhydride	18: benzoic acid
19: (3-aminopropyl)triethoxysilane	20: tetraethoxysilane	21: acetic anhydride
22: 4,4'-(hexafluoroisopropylidene)diphthalic anhydride	23: tetramethoxysilane	24: water
25: 4,4'-oxidianiline	26: 1,4-bis(4-aminophenoxy)-2-tert-butylbenzene	27: pyridine
28: 3,3',4,4'-benzophenonetetracarboxylic dianhydride	29: 1,4-diaminobutane	30: 2,2-bis[4-(4-aminophenoxy)phenyl]hexafluoropropane
31: 2,2'-bis(trifluoromethyl)-4,4'-diaminodiphenyl ether	32: 2,2'-bis(trifluoromethyl)-4,4'-diaminobiphenyl	33: 2,2-bis(3-amino-4-hydroxyphenyl) hexafluoropropane
34: 2-isopropylaniline	35: 4-methylbenzaldehyde	36: hydrazine monohydrate
37: palladium on carbon	38: 1,4-phenylenediamine	39: 1,2,4-triaminobenzene dihydrochloride
40: 4,4'-isopropylidene diphenol	41: anhydrous potassium carbonate	42: phenyltrimethoxysilane
43: ethanol	44: hydrochloric acid	45: dicyclohexylcarbodiimide
46: 4-iodoaniline	47: sodium nitrite	48: hydrogen
49: bicyclo[2.2.2]oct-7-ene-2,3,5,6-tetracarboxylic dianhydride	50: 4,4'-methylenebis(2-methylcyclohexylamine)	51: 1,2,3,4-cyclobutanetetracarboxylic dianhydride
52: 3,4,9,10-perylenetetracarboxylic acid dianhydride	53: 1,12-diaminododecane	54: ethylenediamine
55: o-toluidine	56: formaldehyde	57: tetramethyl bicyclo[2.2.2]oct-7-ene-2,3,5,6-tetracarboxylate
58: thionyl chloride	59: 4-nitro-1,2-phenylenediamine	60: sodium dichromate
61: sulfuric acid	62: sodium hydroxide	63: bisphenol a
64: potassium carbonate	65: 4-fluoro-3-trifluoromethylphenylboronic acid	66: sodium carbonate
67: silicon tetrachloride	68: benzene	69: 4-nitrophenol
70: 4-nitroaniline	71: graphite	72: potassium permanganate
73: concentrated sulfuric acid	74: phosphoric acid	75: potassium chlorate
76: potassium hydroxide	77: 4,4'-methylenebis(cyclohexylamine)	78: phosphorous oxychloride
79: phosphorous pentachloride	80: nitric acid	81: nitrobenzene
82: 2,2,2-trifluoroacetophenone	83: trifluoromethyl iodide	84: phenol
85: chlorine	86: oxygen	87: bromobenzene
88: 1,3-dibromopropane	89: tributyltin hydride	90: 2-(4-nitrophenyl)furan
91: 2,2-bis(4-hydroxyphenyl)hexafluoropropane	92: trifluoromethanesulfonic anhydride	93: diethylamine
94: 2-chlorotrifluoromethylbenzene		

Figure 4. Expanded retrosynthetic pathway tree for polyimide based on 197 articles. For simplicity, duplicate child nodes with the same name at each node were hidden. The number of nodes before expansion was 322 (113 in the figure), and the number of nodes after expansion was 3099 (294 in the figure). The reaction path obtained based on criterion 1 (**Figure 5**) is highlighted in orange. The reaction path obtained based on criterion 2 (**Figure 6**) is highlighted in pink.

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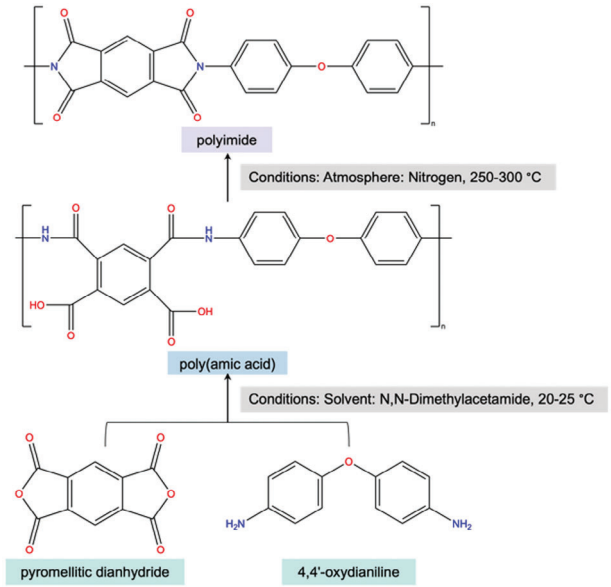

Figure 5. The optimal reaction pathway recommended by the agent is based on criterion 1 (Also see Figure 4).

the discovery of additional alternative pathways to better meet the demands of various practical application scenarios.

4. Discussion

4.1. Challenges and Proposed Solutions in Macromolecule Retrosynthesis Planning

Information extraction for polymer materials presents greater challenges compared to small-molecule chemicals. Small molecules benefit from standardized representations such as Simplified Molecular Input Line Entry System (SMILES)[54,55] and International Union of Pure and Applied Chemistry nomenclature, which provide unique and structured identifiers for molecular structures. In contrast, polymers lack a single, universally recognized naming standard. Their nomenclature often varies based on their naming systems, monomer composition, topology, material properties, application scenarios, and other factors. For instance, poly(vinyl acetate) can be named "poly(1-acetoxyethylene)" or "poly(ethenyl acetate)" based on different nomenclature systems.[11,12] Similarly, polyimides can be named "poly(amide-imide)" or "poly(1,3-dioxoisoindoline-2-yl) ethylene" based on their structural characteristics.[56]

Traditional rule-based methods struggle with these complexities, while LLMs excel at distinguishing various chemical substance names and accurately extracting polymer-related information, without relying on a predefined format. However, due to input length limitations, LLMs ensure consistency only within single articles. To address cross-article inconsistency, the agent reviews the knowledge graph, unifies duplicate nodes, and corrects the graph to achieve entity alignment.

With the help of LLMs and specifically designed techniques, the names of the extracted chemical compounds from different articles were aligned to a great extent, ultimately leading to the completion of the polyimide reaction pathway tree (Figure 4). However, a very small number of duplicate compounds inevitably remained due to the limitations of LLMs, which have been corrected. This minor error highlights the challenges posed by the naming conventions of polymer systems. Furthermore, the large-scale application of our method relies on future improvements in the ability of LLMs to accurately identify chemically identical compounds with different names without omissions.

4.2. Advantages of Using Knowledge Graphs in Macromolecule Retrosynthesis Planning

When processing large volumes of academic literature, traditional Retrieval-Augmented Generation (RAG) techniques[32] are helpful in reducing hallucinations by linking answers to retrieved documents. However, they face significant limitations, including poor document retrieval quality, suboptimal ranking of relevant documents, and unstructured data management.[32,57,58] These shortcomings often lead to incomplete or misleading responses, particularly in retrosynthesis planning, where precision is paramount.

To address these issues, we adopt a structured knowledge graph to store information on chemical reactions from various sources, rather than relying on the vector-based retrieval mechanism in RAG, which typically retrieves information from unstructured text embeddings. By leveraging the knowledge graph, agents can accurately and efficiently retrieve data to construct retrosynthetic pathway trees. This method is highly scalable, allowing agents to explore relevant synthesis literature and extend intermediates to leaf nodes for reactions that cannot be expanded. It also supports dynamic updates by integrating the latest academic papers, effectively mitigating the knowledge update lag in LLMs. Each chemical reaction is paired with a literature reference, addressing issues of hallucination and unverifiability in LLMs. This enhances the accuracy, reliability, and authority of reaction pathway recommendations.

4.3. Enhancing Reliability in Retrosynthesis for Macromolecular Systems

Traditional retrosynthesis approaches, particularly deep learning-based methods, face significant challenges in reliability, interpretability, and applicability to macromolecular systems. Template-free deep learning models for single-step retrosynthesis often exhibit relatively low prediction accuracy (≈40–60%),[7,8] making it difficult to construct complete and valid synthetic pathways. While template-based deep learning methods achieve higher accuracy (≈70–100%),[5,6] their dependence on predefined annotated reaction templates limits their flexibility. Additionally, deep learning models in general lack interpretability, making it difficult to trace the reasoning behind predicted pathways. Furthermore, existing computational methods, including density functional theory, primarily focus on small molecules due to the absence of extensive polymer reaction databases, the complexity arising from the large number of atoms in macromolecules (typically on the order of 10^2–10^6),[59] and the influence of long-range interactions and solvent effects, which make precise calculations particularly challenging.

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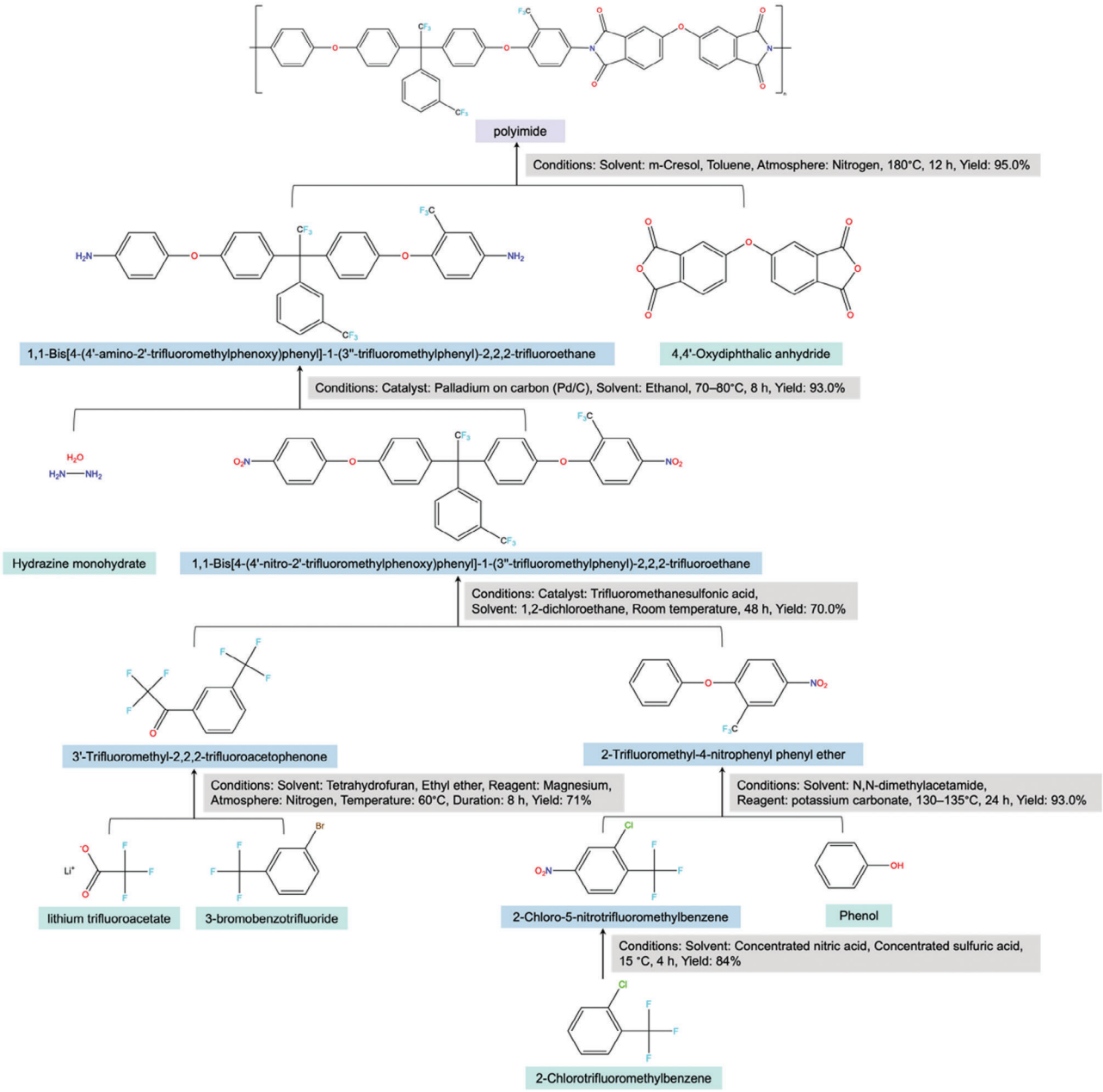

Figure 6. The optimal reaction pathway recommended by the agent is based on criterion 2 (Also see Figure 4).

To overcome these limitations, we propose a novel approach utilizing an LLM agent that leverages authoritative academic literature as its knowledge source to perform multi-step retrosynthesis analysis for polymer materials. Our method enhances reliability and interpretability through multi-level validation. Literature-based validation traces each reaction to authoritative academic sources, minimizing errors and speculation. Database validation aggregates information from sources like eMolecules and PubChem to confirm the commercial availability of intermediates and reactants, ensuring practical feasibility. By integrating experimental findings from literature with comprehensive database verification, our approach provides highly accurate and traceable reaction pathways, achieving estimated accuracy in the high 90 s for complex polymer systems such as polyimides. Furthermore, our method offers multiple viable retrosynthetic routes tailored to different application needs, enhancing its practical value in retrosynthesis planning for macromolecular systems.

4.4. MBRPS: A Novel Approach for Comprehensive Multi-Step Pathway Search

A fundamental limitation of existing retrosynthesis planning methods is their reliance on a "one-to-one" decomposition

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strategy, in which a product is broken down into at most one reaction intermediate. This results in predominantly linear, unbranched pathways that can be efficiently searched using algorithms such as MCTS. However, real-world chemical synthesis often follows a "one-to-many" decomposition strategy, where a single product gives rise to multiple reaction intermediates, forming a multi-branched retrosynthetic network. These multi-branched pathways introduce combinatorial complexity and pose a significant challenge for traditional retrosynthesis algorithms, which are not designed to systematically explore or validate such reaction routes.

To overcome these limitations, we introduce a novel MBRPS algorithm, which systematically traverses retrosynthetic pathway trees using a recursive approach that accounts for both AND and OR relationships between reactant nodes (for detailed implementation, see Section 2.4 of the Experimental Section). It enables the identification of all viable reaction pathways, even in complex multi-branched retrosynthesis scenarios, including reaction conditions. By ensuring that no feasible pathway is overlooked, this method allows for a more exhaustive and logically consistent retrosynthetic search, overcoming the limitations of traditional single-branch decomposition methods.

4.5. Modularity and Integration with Computational Tools

Our retrosynthesis planning agent features a modular architecture that seamlessly integrates with computational tools via API interfaces. We use Python scripts to invoke external tools for literature retrieval, data extraction, retrosynthetic pathway construction, and reaction validation. Key integrations include Google Scholar API for literature search, PyMuPDF for text extraction, RDKit for molecular structure conversion, and PubChem/eMolecules for compound validation. Our retrosynthetic pathway construction employs both existing tools and custom-designed tools, such as MDFS algorithm for pathway tree construction and MBRPS algorithm for pathway searching.

This API-driven, flexible design allows for easy incorporation of additional computational and experimental tools, such as DFT calculations for reaction feasibility assessment and robotic synthesis platforms[60] for automated validation. The system's adaptability ensures seamless integration of evolving computational chemistry methodologies, making it a scalable framework for retrosynthesis planning.

4.6. Future Prospects and Enhancements

Recent breakthroughs in Artificial Intelligence (AI) reasoning models, such as OpenAI's o1 and DeepSeek's R1[61] by leveraging reinforcement learning algorithms for fine-tuning. Additionally, Knowledge graphs play a crucial role by organizing information in a graph structure, improving model reasoning across various domains.[23–26] As these models advance, we can expect more precise and efficient AI systems that will revolutionize fields like drug discovery and materials science, offering innovative solutions to complex problems. Integrating these models into retrosynthesis planning could optimize synthesis strategies.

Leveraging advanced AI reasoning models in combination with constraint-based search algorithms offers a way to constrain simpler precursors to meet specific targets, such as sustainability, safety, or belonging to a particular class of chemical structures. Deep learning techniques, like graph neural networks, can be employed to predict sustainability scores, ensuring more efficient and targeted precursor selection. Sustainability metrics can be derived from databases such as the Sustainable Chemistry Database or EcoScale, which evaluate the environmental impact of chemicals. Additionally, safety data can be queried from the PubChem database, while GHS hazard ratings can be used to exclude dangerous molecules. Further safety and regulatory information can be obtained from TOXNET, TSCA, and IRIS databases, which provide detailed toxicity, environmental, and health risk assessments for chemicals. By integrating these diverse datasets, the system can ensure a more informed and responsible selection of precursors for retrosynthesis, driven by quantifiable criteria such as sustainability scores, GHS hazard classifications, and regulatory compliance information.

5. Conclusion

This study represents the first attempt to develop a fully automated retrosynthesis planning agent specifically designed for macromolecules by integrating LLMs with knowledge graphs. Demonstrated through a case study on polyimide, the approach automates literature retrieval, reaction data extraction, database querying, construction of retrosynthetic pathway trees, further expansion through the retrieval of additional literature on intermediates, and finally searching, evaluation, and recommendation of the optimal route based on conditions, reactants, safety, and other factors. Our approach is versatile and not limited to small molecules but extends to complex macromolecules. In contrast to previous methods that have been limited to a "one-to-one" decomposition strategy, our method is suitable for a "one-to-many" decomposition strategy, a scenario more commonly encountered in practical chemical synthesis analysis. By applying this approach to the widely-used polyimide, the agent successfully constructs the retrosynthesis pathway tree and recommends both established and novel pathways without human intervention. This example demonstrates that with more powerful LLMs, an automated retrosynthesis planning agent could significantly accelerate the discovery of reverse chemical reaction pathways, thereby greatly enhancing research efficiency, especially given the overwhelming volume of materials-related literature that researchers must navigate.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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data mining, knowledge graph, large language model agent, retrosynthesis planning

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